

$$\omega = \frac{-1 + \sqrt{3}i}{2}$$

$$\left(2 \cdot \frac{-1 + \sqrt{3}}{2 + 1} \right)^7$$

$$(2\omega)^7 + (2\omega^2)^7$$

$$= 2^7 \omega^7 + 2^7 \omega^{14}$$

$$= 2^7 (\omega + \omega^2)$$

$$0 = (2 + 2^7) = -128 //$$

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(viii)

(b)

$$x = \frac{1}{2} (-1 + \sqrt{3}) = \omega$$

$$y = \frac{1}{2} (-1 - \sqrt{3}) = \omega^2$$

$$x^3 + y^3 = x^3 + \frac{1}{y^3}$$

$$= \omega^3 + \frac{1}{\omega^6}$$

$$= 1 + 1 = 2$$

$$\omega^3 = 1$$

$$\omega^6 = 1$$

$$(ix) \left(\frac{-1+\sqrt{-3}}{2} \right)^n + \left(\frac{-1-\sqrt{-3}}{2} \right)^n \quad (ix)$$

$$= \omega^n + \omega^{2n}$$

(1) $\rightarrow$

अब, $n=3m$ $\textcircled{1}$, $n \in \mathbb{Z}$ $\textcircled{1}$

$$\textcircled{2} \rightarrow \omega^{3m} + \omega^{6m} = 1 + 1 = 2 \quad \omega = 1$$

$$\therefore n \rightarrow (2)$$

अब, $n=3m+1$, $m \in \mathbb{Z}$

$$\therefore \omega^{3m+1} + (\omega^2)^{3m+1}$$

$$= \omega^{3m} \cdot \omega + \omega^{6m} \cdot \omega^2 = 0$$

$$= \omega + \omega^2 = -1$$

अब, $n=3m+2$, $m \in \mathbb{Z}$

$$\therefore \omega^{3m+2} + (\omega^2)^{3m+2}$$

$$= \omega^{3m} \cdot \omega^2 + \omega^{6m} \cdot \omega^4$$

$$= \omega^2 + \omega = -1$$

$$\frac{(x^i)}{(1+x+x^2)^n} = p_0 + p_1 x + p_2 x^2 + \dots + p_{2n} x^{2n}$$

① मा. व $x = 1, 0, 0$ वास्तविक आहे, — ①

$$z^n = p_0 + p_1 + p_2 + \dots + p_n + p_{2n} \quad \text{--- (2)}$$

$$\begin{aligned} \frac{\mu = \omega}{0} &= 1 + 1 = \\ &= P_0 + P_1 \omega + P_2 \omega^2 + P_3 \omega^3 + P_4 \omega^4 \\ &\quad + P_5 \omega^5 + \dots + P_{2n} \omega^{2n} \\ &= P_0 + P_1 \omega + P_2 \omega^2 + P_3 + P_4 \omega + P_5 \omega^2 \\ &\quad + \dots + P_{2n} \omega^{2n} \end{aligned}$$

$$0 = (p_0 + p_3 + p_6 + \dots) + \omega (p_2 + p_4 + p_7 + \dots) + \omega^2 (p_1 + p_5 + p_8 + \dots)$$

③ $\frac{m}{\omega} + s + m$ $+ (m_s) \omega + s + m$

$$x = \omega^{\sim}$$

$$0 = P_0 + P_1 \omega^{\sim} + P_2 \omega^4 + P_3 \omega^6 +$$

$$P_4 \omega^8 + \dots + P_{2n} \omega^{4n}$$

$$= P_0 + P_1 \omega^{\sim} + P_2 \omega + P_3 + P_4 \omega^{\sim} + P_5 \omega +$$

$$= (P_0 + P_3 + P_6 + \dots) +$$

$$\omega (P_2 + P_5 + P_8 + \dots)$$

$$+ \omega^{\sim} (P_1 + P_4 + P_7 + \dots) \quad \text{--- (4)}$$

$$\textcircled{2} + \textcircled{3} + \textcircled{4} \Rightarrow$$

$$3^n = 3 (P_0 + P_3 + P_6 + \dots) + P_1 (1 + \omega + \omega^{\sim}) + P_4 (1 + \omega + \omega^{\sim}) + P_5 (1 + \omega + \omega^{\sim}) + \dots$$

$$3^n = 3 (P_0 + P_3 + P_6 + \dots)$$

$$\therefore P_0 + P_3 + P_6 + \dots = 3^{n-1}$$